

Bangladesh University of Engineering and Technology

CSE208: Data Structure and Algorithms Sessional II

Online - 4 on Maximum Flow

Section: B

Time: 30 Minutes

Maximum Number of Paths

You are given a communication network with N nodes and M links. Each link can transmit data in only one direction. Your task is to find the **maximum number of possible paths** from a source node s to a destination node t , where **no two paths share a link**.

Input format

- N M
- s t
- Then M lines, each with two integers u v , denoting a link from u to v

Output format

- Print the maximum number of link-disjoint paths. If t is not reachable from s , print -1.

Testcase 1

Input	Expected Output
4 5 0 3 0 1 0 2 1 3 2 3 1 2	2

Explanation

Possible paths from 0 to 3:

1. $0 \rightarrow 3$
2. $0 \rightarrow 2 \rightarrow 3$
3. $0 \rightarrow 1 \rightarrow 2 \rightarrow 3$

Path 1 does not share any link with path 2 and 3, so it can be selected. Path 2 and 3 share a link (2 -> 3), so only one of these two can be selected. Thus, there are two link-disjoint paths from 0 to 3.

Testcase 2

Input	Output
6 7 0 5 0 1 0 2 1 3 2 3 3 5 3 4 4 5	2

Testcase 3

Input	Expected Output
4 5 3 0 0 1 0 2 1 3 2 3 1 2	-1